

~~Q~~ (i) if $f(x) = \frac{x^2 - 3x + 1}{x - 1}$, find $f(-2) + f\left(\frac{1}{3}\right)$

(ii) if $f(x) = \frac{x^2 + 9}{\sqrt{x} - 3}$, find $f(4) + f(5)$

(iii) if $f(x) = x^2 + 3$ then find the value of x if $f(x) = f(2x+1)$

(iv) if $f(x) = \frac{x+5}{3x-4}$ & $t = \frac{5+4x}{3x-1}$ show that $f(t) = x$

Question
Nos.EX-5

If

 $A =$

$$\begin{bmatrix} 2 & 4 & 4 \\ 4 & 2 & 4 \\ 4 & 4 & 2 \end{bmatrix}$$

Show that $A^2 - 8A$ is a scalar matrix.

EX-6find x, y, z if

$$\left\{ \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & 1 \\ 3 & 1 & 2 \end{bmatrix} + 2 \begin{bmatrix} 3 & 0 & 2 \\ 1 & 4 & 5 \\ 2 & 1 & 0 \end{bmatrix} \right\} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

EX-7If $A = \begin{bmatrix} 1 & -3 \\ 2 & -1 \end{bmatrix}$ and

$B = \begin{bmatrix} 1 & 0 & 1 \\ 2 & -3 \end{bmatrix}$ verify that

$$(AB)^T = B^T A^T$$

EX-8

If $A = \begin{bmatrix} 1 & 2 & 6 \\ 7 & 4 & 10 \\ 1 & 3 & 5 \end{bmatrix}$

find A^2 ~~$8A$~~ .

Ex-9 Resolve into partial fractions

$$\frac{1}{(x+1)^2(x+2)}$$

Ex-10 Resolve into partial fractions

$$\frac{2x-3}{(x^2-1)(x+1)} =$$

Ex-11 Resolve into partial fractions

$$\frac{x-5}{x^3+x^2-6x}$$

Ex-12 Resolve into partial fractions

$$\frac{3x-1}{(x-4)(2x+1)(x-1)}$$

Ex-13 If $A = \begin{bmatrix} 1 & 3 & 2 \\ -1 & 2 & 0 \\ 4 & 0 & 3 \end{bmatrix}$

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 1 & 2 \\ 2 & 2 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

Question
Nos.Mark
Obtain

then find matrix D
given that $2A - 2B - D = C$

Ex-14 Find x, y, z if

$$\begin{bmatrix} 2+x & -1 & 3 \\ 0 & y & z \\ 4 & 1 & 3 \end{bmatrix} + \begin{bmatrix} 1+x & 2 & 3 \\ 0 & 1+y & 4 \\ 2 & 3 & 5 \end{bmatrix} = \begin{bmatrix} 6 & 1 & 6 \\ 0 & -1 & 6 \\ 6 & 4 & 8 \end{bmatrix}$$

Ex-15 If $A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 \\ 3 & 0 \\ 0 & 1 \end{bmatrix}$

~~find $(A+B)$~~

find AB .

4 marks

Determinant

Ex:
$$\begin{vmatrix} 1 & x & x^2 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix} \text{ find } x.$$

Ex: Find value of x

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \end{vmatrix} = 0.$$

Ex: Find value of x

$$\begin{vmatrix} 3 & 4 & 3 \\ 5 & x & x \\ 1 & 3 & 2 \end{vmatrix} = 0$$

6 MarksQ1] Cramer's Rule.
(Tutorial-1)

Q2] Partial fractions.

Resolve into partial fraction

1) $\frac{x^2 + 23x}{(x+3)(x^2+1)}$

3) $\frac{3x-2}{(x+2)(x^2+4)}$

2) $\frac{1}{x^3 - 1}$

4) $\frac{x^2 + 36x + 6}{(x-1)(x^2+2)}$

8 MarksQ1] solve these equations
using Matrix Inversion
Method (Tutorial-3)